

NovaFab Systems

Manufacturing Data Engineer - Practice Job Description

DOCUMENT PURPOSE

This is a fictional recruitment document created for PDF parsing, text extraction, LangChain, RAG, chunking, metadata filtering, and question-answering practice. The company, roles, projects, metrics, and people described here are invented.

1. Position Overview

NovaFab Systems builds automation and analytics platforms for advanced manufacturing sites. The Manufacturing Data Engineer connects equipment, quality, maintenance, and production data so engineering teams can make faster decisions.

The role combines data engineering, process engineering, and production support. The engineer works closely with manufacturing, quality, equipment, IT, and analytics teams to create reliable pipelines from shop-floor systems to cloud or on-premise analytical environments.

2. Employment Summary

- Job family: Data Engineering / Smart Manufacturing
- Employment type: Full-time
- Location: Daejeon or Cheonan, Republic of Korea
- Work style: Primarily on-site during ramp-up, hybrid after stabilization
- Travel: Up to 20 percent for new line installation and customer support
- Experience guideline: 2 to 6 years, but strong entry-level candidates may be considered

3. Why This Role Exists

Production teams often own valuable data that is difficult to use because equipment logs, MES records, inspection results, and engineering notes are stored separately. This role creates governed data products that make those sources searchable, traceable, and useful for decision making.

- Reduce time spent manually collecting engineering data
- Improve root-cause analysis with traceable equipment and process histories
- Support SPC, OEE, yield, defect, and maintenance analytics
- Enable AI and machine-learning use cases with reliable training datasets

Core Responsibilities

1. Data Pipeline Development

- Build and maintain batch and near-real-time pipelines for equipment, MES, quality, and maintenance data.
- Design ingestion patterns for CSV, JSON, REST APIs, relational databases, message queues, and machine logs.
- Create transformation logic that standardizes timestamps, equipment IDs, lot IDs, units, and process names.
- Implement data validation rules for missing values, duplicate records, schema drift, and abnormal ranges.
- Document source-to-target mappings and operational dependencies.

2. Manufacturing Data Modeling

Create analytical tables that allow users to trace production by lot, unit, equipment, recipe, shift, and inspection result. Models should support both engineering analysis and dashboarding.

- Fact tables for process events, alarms, measurements, defects, and downtime
- Dimension tables for equipment, product, process, recipe, operator group, and calendar
- Slowly changing dimensions when equipment configuration or recipe definitions change
- Clear ownership and data lineage for critical production metrics

3. Reliability and Operations

- Monitor pipeline freshness, throughput, and failure rates.
- Add logging, retry logic, alerts, and runbooks for common failures.
- Support incident response when data latency affects engineering or production decisions.
- Perform root-cause analysis and implement preventive actions after repeated failures.

4. Cross-functional Collaboration

- Translate process-engineering questions into data requirements.
- Review dashboard definitions with quality and production teams.
- Work with IT and security teams on account, network, and access-control requirements.
- Explain data limitations clearly instead of hiding uncertainty.

Technical Requirements

Required Skills

- Python for data processing, automation, API interaction, and testing.
- SQL including joins, window functions, aggregation, subqueries, and query optimization basics.
- Pandas or Polars for exploratory data analysis and transformation.
- Git-based version control and pull-request workflows.
- Basic Linux command-line usage and virtual-environment management.
- Understanding of REST APIs, JSON, CSV, Parquet, and relational databases.
- Ability to debug data issues systematically using logs and small reproducible tests.

Preferred Skills

- Airflow, Dagster, Prefect, or another workflow orchestration platform.
- Spark, DuckDB, dbt, Kafka, or cloud data platforms.
- FastAPI for lightweight internal data services.
- Streamlit for engineering tools and prototypes.
- Docker and basic CI/CD experience.
- Experience with manufacturing systems such as MES, SPC, historian, PLC, SCADA, or equipment interfaces.
- Exposure to LangChain, vector databases, RAG, embeddings, or document-processing workflows.

Engineering Foundations

Candidates do not need to be experts in every manufacturing domain. However, they should be comfortable learning process terminology and connecting business questions to measurable data.

- Yield, defect rate, cycle time, takt time, OEE, downtime, and throughput
- Basic statistical concepts such as mean, standard deviation, quartiles, correlation, and process capability
- Difference between correlation, causation, and operational coincidence
- Importance of traceability and reproducibility in production analysis

Example screening question: How would you determine whether a sudden yield drop is caused by one piece of equipment, one recipe revision, or one incoming material lot?

Sample Project A - Equipment Log Pipeline

Business Context

A production line generates text logs from 120 machines. Engineers currently copy selected files into spreadsheets when a problem occurs. The files contain timestamps, equipment states, alarms, setpoints, measured values, and operator actions.

Target Architecture

- Collect raw logs every five minutes from controlled network locations.
- Parse lines into structured records and preserve the raw source for auditability.
- Normalize timestamps to a single timezone while retaining original timestamps.
- Write structured data into partitioned storage by date and equipment.
- Create curated tables for alarm events, recipe changes, and process measurements.
- Expose a query layer for engineering notebooks and dashboards.

Data Quality Rules

- Reject records with invalid equipment identifiers.
- Flag but retain rows with out-of-range sensor values.
- Detect duplicate event IDs within the same source file.
- Compare expected and actual file counts by hour.
- Record parser version so reprocessing remains traceable.

Success Metrics

- Data freshness under 10 minutes for 95 percent of normal operating periods.
- Parsing success rate above 99.5 percent.
- Root-cause analysis preparation time reduced from hours to minutes.
- Zero loss of original source files during normal retention period.

Example Interview Follow-up

If the schema of one equipment family changes without notice, explain how you would detect the change, prevent silent corruption, preserve the raw data, and restore the curated dataset.

Sample Project B - Quality Analytics

Scenario

A final inspection process measures width, thickness, alignment, and surface defects. The quality team wants a daily report that highlights drift before specification failures increase.

Available Data

- Inspection measurements at unit level
- Specification limits by product code and revision
- Equipment and tool identifiers
- Recipe version
- Production timestamp and shift
- Rework and scrap classification
- Maintenance events and calibration history

Expected Analysis

- Calculate mean, standard deviation, median, quartiles, minimum, and maximum by product and day.
- Track specification violations and warning-band violations.
- Calculate process capability indicators only when the data and assumptions are appropriate.
- Compare distributions before and after recipe or equipment changes.
- Identify whether apparent improvement is caused by product mix changes.

Example SQL Task

Create a query that returns daily average thickness, standard deviation, unit count, and defect count by equipment for the most recent 30 production days. Exclude engineering test lots but do not exclude reworked production units unless the business rule explicitly requires it.

Common Mistakes to Avoid

- Using only averages and ignoring distribution shape.
- Comparing different product families without normalization.
- Treating missing measurements as zeros.
- Calculating capability metrics without checking specification definitions and process stability.
- Assuming a dashboard trend proves the cause of a process change.

Sample Project C - Internal RAG Assistant

Use Case

Manufacturing engineers maintain setup guides, troubleshooting notes, equipment manuals, quality procedures, and issue reports. The goal is to create an internal question-answering assistant that retrieves relevant passages and shows source references.

Document Processing Flow

- Collect approved PDF and DOCX documents from controlled repositories.
- Extract text and metadata such as document title, revision, department, effective date, and page number.
- Chunk documents using a strategy that preserves section meaning instead of splitting only by fixed character count.
- Create embeddings and store them with metadata in a vector database.
- Retrieve candidate chunks, optionally rerank them, and provide them to the language model.
- Return an answer with citations to the original document and page.

Evaluation Criteria

- Retrieval recall: relevant source text should appear in the retrieved set.
- Answer grounding: the generated answer should be supported by retrieved content.
- Citation accuracy: source references must point to the passage actually used.
- Freshness: obsolete document revisions should not be preferred over active revisions.
- Security: users should retrieve only documents they are allowed to access.

Example Chunk Metadata

document_id = SOP-ETCH-014; revision = 7; page = 12; section = Chamber Cleaning; equipment_family = EF-300; effective_date = 2026-03-01; confidentiality = Internal.

Practice question: Why is storing page number and document revision useful in a RAG system? Answer using the evaluation criteria above.

Data Engineer Coding Exercise

Input Dataset

You receive a CSV file named `process_measurements.csv` with the columns below. The file may contain missing values, duplicate rows, and multiple units of measurement.

Column	Example	Meaning
timestamp	2026-09-18 09:15:00	measurement time
equipment_id	EQ-014	equipment identifier
lot_id	L260918A	production lot
metric	thickness	measurement name
value	42.7	numeric measurement
unit	um	unit of measurement
recipe	RCP_21B	recipe version

Task 1 - Cleaning

- Parse timestamp as datetime.
- Remove exact duplicate rows.
- Convert thickness values from mm to um when required.
- Report rows that cannot be converted safely.
- Do not silently replace missing measurement values with zero.

Task 2 - Aggregation

- For each equipment and date, calculate count, mean, standard deviation, Q1, median, Q3, minimum, and maximum.
- Return only groups with at least 20 valid measurements.
- Sort by date and equipment ID.

Task 3 - Explainability

Write two or three sentences describing what additional information you would need before deciding whether an unusual distribution represents a real process problem.

Behavioral and Collaboration Interview

Question 1 - Ambiguous Requirement

A process engineer asks for a dashboard showing bad equipment. The phrase bad equipment has no agreed definition. Describe how you would clarify the requirement, define metrics, and prevent the dashboard from becoming misleading.

Question 2 - Production Incident

A pipeline fails during a production issue. Engineering wants immediate data, while IT asks you to follow the normal change process. Explain how you would prioritize safety, data integrity, speed, and communication.

Question 3 - Disagreement About Data

Two teams report different yield numbers for the same day. One uses completion timestamp and the other uses start timestamp. Describe how you would identify the difference and create a governed definition.

Question 4 - Learning New Technology

Describe a time you had to learn a new tool quickly. A strong answer should explain the initial gap, the learning strategy, a small validation step, and how the new capability was applied to real work.

Question 5 - Communication

Explain a technical data-quality issue to a manager who does not need implementation details. Focus on business impact, current containment, confidence level, and next action.

What Interviewers Look For

- Structured problem solving rather than immediate guessing
- Clear separation of facts, assumptions, and hypotheses
- Appropriate use of small tests before large-scale changes
- Ownership without hiding uncertainty
- Respect for production controls and cross-functional constraints

Glossary for Manufacturing Data

MES - Manufacturing Execution System. Tracks production execution, lots, routes, operations, and related records.

PLC - Programmable Logic Controller. Industrial controller used to operate equipment and automation.

SPC - Statistical Process Control. Methods used to monitor process behavior and variation over time.

OEE - Overall Equipment Effectiveness. A composite metric related to availability, performance, and quality.

Yield - Share of output meeting the defined acceptance criteria. The exact denominator and timing rule must be governed.

Traceability - Ability to connect output to relevant process, equipment, material, recipe, and inspection history.

Schema drift - Unexpected change in source fields, types, or structure that can break or silently alter pipelines.

Data lineage - Record of how data moves and transforms from source to final analytical output.

Idempotency - Property that allows a repeated pipeline run to produce the same intended result without duplicate side effects.

Backfill - Reprocessing historical periods after fixing logic or adding newly required fields.

Embedding - Vector representation used to compare semantic similarity between pieces of text or other content.

RAG - Retrieval-Augmented Generation. A pattern in which relevant external content is retrieved and supplied to a language model.

Chunk - A retrievable section of a larger document. Chunk design affects context quality and search recall.

Metadata filter - A retrieval constraint such as document revision, department, date, equipment family, or security level.

Hallucination - Generated content that is unsupported or incorrect relative to the available evidence.

Practice retrieval task: Find the definitions of schema drift, data lineage, idempotency, and metadata filter. Then explain how they could interact in one production data system.

Final Practice Questions

Document Search Questions

- What is the target data freshness for the equipment log pipeline?
- Which systems are named as potential manufacturing data sources?
- Why should raw equipment logs be preserved even after parsing?
- What metadata fields are suggested for the RAG assistant?
- What minimum group size is required in the coding exercise aggregation?
- Which interview question deals with inconsistent yield definitions?
- How does the document define idempotency?
- What are three common mistakes in quality analytics?

Reasoning Questions

- Design a chunking strategy for this PDF that would work better than fixed 500-character chunks.
- Propose metadata fields that would allow retrieval by section, page, topic, and project.
- Explain when overlap between chunks may help and when it may create redundant retrieval results.
- Describe how you would test whether a PDF loader preserved page numbers correctly.
- Explain the difference between retrieving the right passage and generating a correct answer.

Mini Assignment

Load this PDF with a PDF loader of your choice. Print the number of pages, total extracted characters, and the first 300 characters from page 1. Then split the document, create embeddings, store them in a vector database, and ask at least five questions whose answers appear on different pages.

Suggested Validation Output

- Page count should be 10.
- Extracted character count should be clearly greater than zero.
- Page metadata should remain available after loading.
- At least one question should retrieve content from page 4 or later.
- Citations should identify the page used for the answer.

End of practice document